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(*c*) Consider a convolutional encoder defined over $GF(4)$ with the generator polynomials

$$g_1(D) = 2D^3 + 3D^2 + 1$$

$$g_2(D) = D^3 + D + 1$$

(*i*) What is the minimum distance of this code?

(*ii*) Is this code non-catastrophic? Why? 7

(*d*) Explain in brief about Viterbi decoding of convolutional codes. 7

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BE (6th Semester)

Examination, April - May, 2016

(New Scheme)

Information Theory and Coding

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Attempt all five units. Part (*a*) of each unit is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*) of each unit.

(*ii*) The figures in the right-hand margin indicate marks.

Unit-I

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1. (*a*) A source emits one of four possible messages m_1, m_2, m_3 and m_4 with probabilities 1/2, 1/4, 1/8, 1/8 respectively. Calculate the information content of the message. 2

(*b*) Explain in brief about various types of communication channels. 7

(*c*) What do you mean by source coding? Also explain code length and code efficiency. 7

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(*d*) What are the important factors that can affect the choice of line coding? 7

Unit-II

2. (*a*) What is the difference between static and dynamic Huffman coding? 2

(*b*) Construct the Huffman code tree for the following message where value of $M=3$: 7

$[x] = [\, x_1 \;\; x_2 \;\; x_3 \;\; x_4 \;\; x_5 \;\; x_6 \;\; x_7 \,]$
$[p] = [0.4 \;\; 0.2 \;\; 0.12 \;\; 0.08 \;\; 0.08 \;\; 0.08 \;\; 0.04]$

(*c*) A black and white TV picture consists of 52 lines of picture information. Assume each line consists of 525 picture elements and each element has 256 brightness levels. Pictures repeated at the rate of 30 per second. Calculate average rate of information content. 7

(*d*) A Gaussian channel has 1 MHz bandwidth. Calculate the channel capacity if the signal power to noise spectral density ratio is 10^5Hz. Also find maximum information rate. 7

Unit-III

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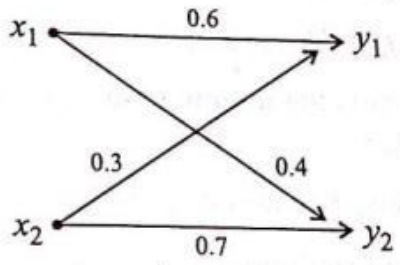
3. (*a*) What is Binary symmetric channel? 2

(*b*) Explain in brief about S/N trade off. 7

(*c*) Consider an AWGN channel with 4 kHz bandwidth and the noise power spectral density $\eta/2 = 10^{-12}$ W/Hz. The signal power required at the receiver is 0.1 mW. Calculate the capacity of this channel. 7

(*d*) Find channel capacity and mutual information shown in figure, where $p(x_1) = 0.3$ and $p(x_2) = -0.7$. 7



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Unit-IV

4. (*a*) What do you mean by double error detection? 2

(*b*) Explain how hamming distance can be used for error detection and correction. 7

(*c*) The generator matrix for a (7, 4) block code is given. If the received code word is given by

$R = [\, 1\;1\;1\;1\;1\;0\;0 \,]$

find transmitted code word. 7

(*d*) The generator polynomial of a (7, 4) cyclic code is $g(x) : 1 + x + x^2$. Find all the code words of this code. 7

Unit-V

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5. (*a*) What are turbo encodes and decoders? 2

(*b*) Explain in brief about recursive convolutional encoders. 7